

The Performance of Pre-Founding Entrepreneurial Teams: The Importance of Learning and Leadership

Kristin Knipfer*

Technische Universität München, Germany

Emanuel Schreiner

University of Nebraska at Omaha, USA

Ellen Schmid

Technische Universität München, Germany

Claudia Peus

Technische Universität München, Germany

Entrepreneurial teams often struggle with simultaneous task and team challenges at an early stage of new venture creation. The way in which teams shape their teamwork is key in leveraging performance in the pre-founding phase. Learning should help the team in establishing good teamwork and in expanding its members' entrepreneurial capabilities. Leadership is needed to facilitate and guide this learning. Accordingly, we investigated learning and leadership as facilitators of performance in the pre-founding phase. Specifically, we examined team reflexivity as a collective internal learning process and boundary spanning behaviour as an externally directed individual activity, operating at different levels in fostering team and individual performance. Charismatic team leadership was examined as a catalyst of learning, shaping team and individual performance ultimately. The multilevel mediation model was tested based on data from 196 members of 58 teams of a venture creation programme. Team reflexivity predicted team and individual performance. Boundary spanning behaviour was not related to performance. As hypothesised, charismatic team leadership predicted team and individual performance, both mediated by team reflexivity. This research highlights the relevance of team learning in pre-founding teams and emphasises leadership in shaping learning and moving new ventures forward.

* Address for correspondence: Kristin Knipfer, TUM School of Management, Technische Universität München, Arcisstrasse 21, 80333 Munich, Germany. Email: kristin.knipfer@tum.de

The majority of new ventures are founded by teams, putting the team at the heart of new venture creation (Schjoedt & Kraus, 2009a). The pre-founding phase—a phase that is qualitatively distinct from later phases and crucial for venture success—puts particularly high demands on the team (Fern, Cardinal, & O'Neill, 2012; Foo, Sin, & Yiong, 2006; McMullen & Shepherd, 2006). Pre-founding teams often struggle with simultaneous task and team challenges (DeMol, Khapova, & Elfring, 2015). A pre-founding team needs to select the most promising idea and refine the business model while changes in all facets of the nascent venture are commonly occurring; at the same time, it needs to establish team functioning and expand its members' entrepreneurial capabilities (Ardichvili, Cardozo, & Ray, 2003). This raises the important question how teams manage to perform in the light of the challenges associated with the pre-founding phase. The literature has seen a surge in studies exploring the success factors of entrepreneurial teams (Frese, Chell, & Klandt, 2000; Ireland & Webb, 2007; Klotz, Hmieleski, Bradley, & Busenitz, 2014). The dominant view highlights the significance of input factors such as team composition and prior experience (Jin, Madison, Kraicz, Kellermanns, Crook, & Xi, 2017). However, the way in which a team shapes its teamwork seems crucial in leveraging performance in the pre-founding phase (Bird, Schjoedt, & Baum, 2012; Cohen & Bailey, 1997). We put forward the assumption that *learning* of the whole team and of its members will foster performance and that *leadership* can facilitate and guide this learning.

Learning should facilitate the development of internal team functioning in the often chaotic course of action and the enhancement of the team members' entrepreneurial capabilities in seeking external support and presenting the endeavour to stakeholders (Keith, Unger, Rauch, & Frese, 2016; Marrone, Tesluk, & Carson, 2007). Hence, we explore the contribution of internal and external learning processes, operating at different levels in fostering team and individual performance (Blackburn & Kovalainen, 2009; Davidsson & Wiklund, 2001): (a) team reflexivity as a collective, internal learning process that is characterised by continuous joint evaluation and discussion of team goals and strategies (Schippers, Den Hartog, & Koopman, 2007), and (b) individual boundary spanning behaviour as an externally directed activity that is performed by each team member to gather expertise and support and to represent the business to stakeholders (Ancona & Caldwell, 1992). Given the relevance of learning, it is important to elucidate the leadership behaviours that can initiate, facilitate, and guide learning in the light of the challenges and uncertainties associated with the pre-founding phase (Gorgievski & Stephan, 2016). As an entrepreneurial undertaking is a *weak situation* involving high ambiguity, uncertainty, and the need for orientation, we see the need for a lead entrepreneur who communicates a clear vision for the joint endeavour and demonstrates confidence in the team members' capability to achieve this vision (Ensley, Pearce, & Hmielski, 2006). Charismatic team leadership projects this picture (Jacobsen & House, 2001; Shamir & Howell, 1999). We thus investigate

charismatic team leadership as a catalyst of learning, shaping the performance of the team and of its members.

Our research advances our understanding of the factors that foster performance in the pre-founding phase—a phase that has been neglected by previous research (Johnson, Parker, & Wijbenga, 2006). By exploring the role of team reflexivity and boundary spanning behaviour, we offer a nuanced picture of how learning enables performance of pre-founding teams. Moreover, we provide new insights into the leadership behaviours that encourage and guide learning. Our findings may help pre-founding entrepreneurial teams to shape teamwork in an effort to reduce the risk of early failure, educators to design entrepreneurship training with impact, and stakeholders to recognise high-potential teams.

THEORY AND HYPOTHESES

The pre-founding phase puts heightened demands on the team as a whole and on each individual team member (McMullen & Shepherd, 2006; Ravasi & Turati, 2005). We think of a pre-founding team as two or more persons who are committed to develop their business model, whose actions reflect the interdependent and interactive nature of their teamwork, and who share the responsibility and accountability in the creation of the future venture (Schjoedt & Kraus, 2009b). These teams face a high level of uncertainty, a lack of standard procedures and work routines, and little structure in their teamwork. This is particularly challenging as they operate in a dynamic context that is characterised by rapidly changing circumstances, high competition, and a need for out-of-the-box-thinking (McMullen & Shepherd, 2006). As a consequence, entrepreneurial teams are required to quickly adapt their goals and strategies whenever necessary and to develop their members' entrepreneurial capabilities in the course of action (Cope, 2003). In other words, the challenges of the pre-founding phase require constant learning, both as a team and as a member of the team (Ravasi & Turati, 2005). Accordingly, learning of the team and its individual members should leverage the performance of pre-founding teams to a large extent. Moreover, lead entrepreneurs will shape team processes and team outcomes in the pre-founding phase to a large extent because of their great autonomy and high discretion in structuring teamwork (Cogliser & Brigham, 2004). Yet, not all entrepreneurs lead the same way, and this variance in leadership is expected to explain variance in processes and outcomes of pre-founding teams. Accordingly, we examined the contribution of *learning* and *leadership* in enhancing the performance of pre-founding teams.

Team-Reflexivity as Team-Level Driver of Entrepreneurial Performance

Pre-founding teams are required to critically evaluate their vision for the new venture on a regular basis and constantly adapt their strategies to achieve this

vision in order to function as a team (West, 2000). Therefore, we propose team reflexivity, “the extent to which team members collectively reflect upon the team’s objectives, strategies, and processes (...) and adapt them accordingly” (West, 2000, p. 3), to be a driver of entrepreneurial performance operating on the team level. Although reflexivity can refer to individual cognition, and individual reflexivity may contribute to team reflexivity (Schipper et al., 2007), it does not represent a team property unless it manifests in shared, social cognition (West, 2000). Therefore, in line with earlier notions, our conception of team reflexivity refers to a group process and includes collective behaviours such as group discussion (Otte, Konradt, Garbers, & Schippers, 2017).¹

A growing number of studies link team reflexivity to team performance in contexts where innovation is required and the environment changes frequently (DeDreu, 2002; Hoegl & Parboteeah, 2006). Team reflexivity is particularly important for teams who are still forming because it includes evaluation and discussion processes that foster team development (Schipper et al., 2007). It ensures that goals and means to achieve these goals are monitored, regularly reviewed, and adapted to changes in the environment; it further allows for effective internal communication and coordination (Shin, Kim, & Lee, 2016). We expect team reflexivity to help entrepreneurial teams in creating working conditions, identifying and initiating tasks at hand, establishing shared routines and team structures, and aligning individual and team goals in pursuing their business idea. It is thus assumed to enhance a team’s performance.

Hypothesis 1: Team reflexivity is positively related to entrepreneurial team performance.

Boundary Spanning Behaviour as Individual-Level Driver of Entrepreneurial Performance

While team reflexivity represents the internal learning of the team as a whole, team members’ *boundary spanning behaviour* describes individual externally directed activities. As the distribution of responsibilities and task work are unclear and fluctuating in the pre-founding phase (Ensley et al., 2006), each individual team member’s boundary spanning efforts cannot be expected to converge. Instead, the choice to interact with external actors is individual in nature and the primary source of variation is on the individual level (Marrone et al., 2007). In our research and in line with previous notions, the agent of

¹ In previous research, team reflexivity was not related to individual reflexivity, thus confirming that both are distinct processes (Schipper et al., 2007). Still, individual and team learning are likely to be interrelated, and models of learning include multiple levels (e.g. Crossan, Lane, & White, 1999).

boundary spanning behaviour is the individual member of a team who interacts with future customers, potential investors, and other teams (Ancona & Caldwell, 1992).

Individual boundary spanning behaviour is particularly important in the pre-founding phase (Gielnik, Krämer, Kappel, & Frese, 2014) as crossing team boundaries shapes team members' reputation and develops their persuasion and impression-management skills (Baron & Tang, 2009). It allows team members to get feedback early, which in turn influences their entrepreneurial capabilities (Leung, Zhang, Wong, & Foo, 2006). Furthermore, it bolsters the team members' contribution to team tasks and their contentment with team membership (Marrone et al., 2007). Individual boundary spanning behaviour is thus assumed to facilitate the performance of each team member.

Hypothesis 2: Boundary spanning behaviour is positively related to the performance of individual members of the entrepreneurial team.

Cross-Level Effects of Team Reflexivity on Individual Performance

Individual outcomes may vary due to team-level properties (Braun, Peus, Weisweiler, & Frey, 2013; Mathieu & Taylor, 2007). Reflexive teams have a more profound understanding of their goals, thereby enabling their members to proactively pursue the team goals (West, 2000). Team reflexivity is considered to create a "climate of task excellence" that encourages team members to excel in task fulfilment (West, 1996, p. 570). Accordingly, we assume that team reflexivity not only influences team performance but also individual performance.

Hypothesis 3: Team reflexivity is positively related to the performance of individual members of the entrepreneurial team.

The Significance of Charismatic Team Leadership for Team and Individual Processes

Leadership can be defined as the "intentional influence by one person over others in order to guide, structure and facilitate . . . activities and relationships" (Yukl, 2002, p. 7); it has been recognised as a key factor for entrepreneurial performance (Ensley, Carland, & Carland, 2000; Hmieleski & Ensley, 2007). Leadership may facilitate team processes and outcomes through influential acts in terms of communicating a strong vision for the joint venture and empowering every team member to follow this vision (Baum, Locke, & Kirkpatrick, 1998; Cardon, Zietsma, Saporito, Matherne, & Davis, 2005). Charismatic leadership describes these behaviours (Jacobsen & House, 2001). It was

conceptualised around three main components: (a) sensitivity to opportunities and constraints in the environment; (b) shaping respectful and positive social relationships within the team through consideration of followers' needs; and (c) empowering them to implement the shared vision through exemplary acts at high personal cost (Conger, Kanungo, & Menon, 2000). In our study, we adopt previous team-level conceptualisations of charismatic leadership (e.g. Agle, Nagarajan, Sonnenfeld, & Srinivasan, 2006; Nohe, Michaelis, Menges, Zhang, & Sonntag, 2013). Entrepreneurial teams are characterised by interdependence in task completion, interactivity in decision-making, and high visibility of the leader. Therefore, although leadership is generic with regard to its application to individuals, teams, or organisations (Kozlowski & Klein, 2000; Kozlowski, Mak, & Chao, 2016; Wang & Howell, 2010), we think of charismatic leadership as a team emergent state, where perceptions converge over time (Fulmer & Ostroff, 2016; Klein & House, 1995).

Charismatic leadership should motivate teams to constantly challenge the status quo (Schipper, Den Hartog, Koopman, & van Knippenberg, 2008) while creating feelings of control and reducing perceptions of threat (Conger et al., 2000). This likely increases trust in the team and the leader (Braun et al., 2013). As a consequence, team members feel safe to speak up and experiment with new approaches and feel comfortable in critically discussing the team's goals and strategies (Paulsen, Maldonado, Callan, & Ayoko, 2009). Charismatic team leadership is thus assumed to foster team reflexivity.

Hypothesis 4: Charismatic team leadership is positively related to team reflexivity.

Charismatic team leadership is expected to enhance team efficacy; this feeling of team effectiveness improves a team's capacity to perform eventually (e.g. Somech, 2006). Based on our premise that team reflexivity supports team performance, we hypothesise an indirect effect of charismatic team leadership on team performance. In addition, charismatic team leaders provide clear direction and encourage followers' initiative (Tesluk & Mathieu, 1999). Feelings of empowerment put forward by charismatic team leadership enable team members to achieve the high performance expectations set by their team leader (Shamir, House, & Arthur, 1993). Based on the above reasoning of the positive effects of charismatic leadership on team reflexivity and the assumed relationship of team reflexivity with individual performance, we propose a cross-level effect of charismatic team leadership on individual performance through team reflexivity.

Hypothesis 5: Charismatic team leadership is positively related to entrepreneurial team performance and this relationship is mediated by team reflexivity.

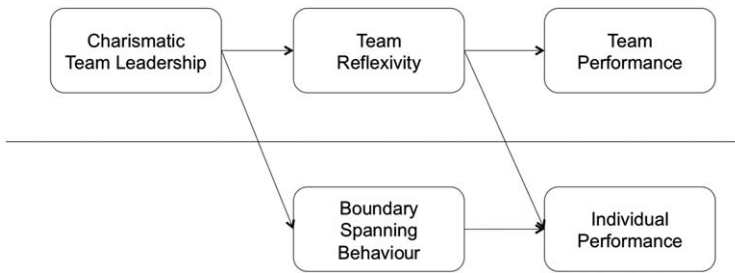


FIGURE 1. Multilevel mediation model of team and individual performance. The upper part describes team-level relationships, the lower part individual-level relationships.

Hypothesis 6: Charismatic team leadership is positively related to the performance of individual members of the entrepreneurial team and this relationship is mediated by team reflexivity.

By putting emphasis on each follower's contribution to the team endeavour, charismatic team leadership creates a "feeling of making a difference" (Conger et al., 2000, p. 752), heightening followers' commitment to act in pursuit of the team goals. Moreover, confidence in their followers' capacity to cross team boundaries is raised (Shea & Howell, 1999). We thus propose that charismatic team leadership fosters boundary spanning behaviour. Based on our assumption that boundary spanning behaviour enhances individual performance, we hypothesise an indirect effect of charismatic team leadership on individual performance that is mediated by boundary spanning behaviour.

Hypothesis 7: Charismatic team leadership is positively related to boundary spanning behaviour of members of the team.

Hypothesis 8: Charismatic team leadership is positively related to the performance of individual members of the entrepreneurial team and this relationship is mediated by boundary spanning behaviour.

Summing up, we propose a multilevel model, in which team reflexivity and individual boundary spanning behaviour facilitate the performance of the whole team and of its members (see Figure 1). Charismatic team leadership is expected to foster these learning processes at multiple levels, influencing entrepreneurial teams to perform better.²

² Instead of testing a research model that theorizes multilevel homology in terms of parallel constructs at multiple levels and parallel functional mechanisms on all levels (Hollingshead & Poole, 2012), we investigate two drivers of entrepreneurial performance at two levels: team reflexivity on the team level and boundary spanning behaviour on the individual level.

METHOD

Sample and Procedure

We gathered data in collaboration with an innovation and start-up centre in Germany. We studied teams who participated in a *venture creation programme* that combines education and incubation (Lackéus & Williams Middleton, 2015), which often results in nascent behaviour (Bae, Qian, Miao, & Fiet, 2014; Martin, McNally, & Kay, 2013). The programme is offered on a regular basis and advertised in the local community and beyond. Participants signed up individually. At the beginning, they met at the *idea fair*, formed teams, and nominated a team leader. The team leader had specific responsibilities such as coordinating the team's activities and reporting its progress to the programme instructors. After four months of weekly training and teamwork, the teams submitted a business plan and presented their business model following the structure of an elevator pitch. Before the evaluation of the business plan and the business pitch were made available, we administered a survey to 252 participants. The participants were asked to sign a consent form in which they granted access to their performance evaluations.

For data analysis, we excluded 17 teams because not all members of the team granted access to their performance evaluations or fewer than two team members provided leadership ratings. The final sample included 196 participants in 58 teams of three to five members (41.4% with three and 53.4% with four members). Participants (23.5% female) were on average 23.8 years old ($SD = 3.2$). Their background was mainly in computer science (25.5%), engineering (19.7%), and business administration (19.4%). Twenty-six per cent of the teams pursued their business idea after finishing the programme by taking part in advanced training and mentoring, participating in national business plan competitions, or nascent behaviours such as applying for funding.

Measures

Team Reflexivity. In line with previous notions (Schippers et al., 2007; West, 2000), we assessed team reflexivity using a referent-shift consensus model (i.e. the team as agent of learning; Chan, 1998). Because all members of the team assessed the same group phenomenon, they were expected to agree on their individual evaluations (Van Mierlo, Vermunt, & Rutte, 2009). All team members, including the lead entrepreneur, responded to eight items that were based on the measure provided by Schippers et al. (2007; see Appendix). Validity and reliability of the original measure had been established by previous research (Schippers et al.,

2007), and short measures of team reflexivity had been used before (e.g. Schippers et al., 2008; Schippers, West, & Dawson, 2015). We selected items to cover both the evaluation and the discussion aspects of team reflexivity.³ The items were provided in the past tense. Team members rated the items on a Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Individual-level reliability was Cronbach's $\alpha = .84$. We calculated r_{wg} , $ICC(1)$, and $ICC(2)$ as indicators for within-team agreement and between-team variance. The mean r_{wg} was .77 (*Median* = .87), which represents strong agreement (LeBreton & Senter, 2008). The $ICC(1)$ was .16, $F(1, 57) = 1.67$, $p < .01$, and the $ICC(2)$ was .40. We conclude that aggregation of individual ratings was legitimate as there was high within-team agreement ($r_{wg} > .70$) and between-team variability (significant F statistics).⁴ Team-level reliability for team reflexivity was Cronbach's $\alpha = .85$.

Boundary Spanning Behaviour. Following our theoretical conceptualisation, we operationalised boundary spanning behaviour on the individual level. We chose four of the twelve original items by Ancona and Caldwell (1992) to assess ambassador activities and three of the four items to assess scout activities of each individual team member, excluding the lead entrepreneur (see Appendix). In case multiple items used repetitive wording or referred to the same content, we selected items that were simple

³ To examine the validity of our short measure, we recruited a comparable sample from the same entrepreneurial training and coaching programme ($n = 20$ teams). In this sample, we used the 18-item team reflexivity scale provided by Schippers et al. (2007, study 2). Examination of item-scale correlations suggested deleting the reverse-coded item "Problems are discussed only once they have become critical." Individual-level Cronbach's α for the full scale was .90. Our 8-item aggregated composite measure correlated very highly and significantly with the 17-item aggregate measure, $r = .91$, $p < .001$. In other words, the shortened measure consisting of eight items still represents 83 per cent of the core variance of the construct.

⁴ Although $ICC(2)$ values were lower than the traditional guidelines (e.g. LeBreton & Senter, 2008), they were similar to values in studies of leadership (e.g. Srivastava, Bartol, & Locke, 2006) and team reflexivity (e.g. Shin et al., 2016). Low ICC values can mask relationships on the team level and, thus, our tests may be considered as conservative tests of the hypothesized relationships (Srivastava et al., 2006). Further, they reflect the small team size (Klein & Kozlowski, 2000). In comparison, the average $ICC(2)$ values ranged from .52 to .56 ($SD = 0.23$ to 0.26) in a recent simulation study (Woehr, Loignon, Schmidt, Loughry, & Ohland, 2015). The authors showed that even low $ICC(1)$ and $ICC(2)$ values effectively differentiate real teams from pseudo teams. Moreover, because entrepreneurial teams are the subject of our research, it is conceptually important to examine team-level effects. We conclude that it was appropriate to aggregate team members' ratings to the team level.

and unambiguous.⁵ Previous research has also reduced the measurement of boundary spanning behaviour to four to six items (Hirst & Mann, 2004; Marrone et al., 2007). The items were adapted to match the setting of the study and reference to an organisational context was omitted. All items were provided in the past tense. Participants rated these items on a Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Individual-level reliability was Cronbach's $\alpha = .80$.

Charismatic Team Leadership. Consistent with our theoretical conception of charismatic leadership as a team construct, we assessed it using a referent-shift consensus model (Chan, 1998), where individual team members were asked to evaluate the behaviour of the team leader towards the whole team (the level of inference is the team; Bliese, 2000). The key theoretical construct is the within-group consensus on the team leader's behaviour towards the collective. The shared perception of charismatic leadership is expected to operate on the team level and to exert a stronger influence than the individual perception (DeGroot, Kiker, & Cross, 2000).

Team members evaluated their lead entrepreneur's leadership behaviour towards the whole team by means of a composite score of five items from the Conger-Kanungo scale of charismatic leadership (Conger et al., 2000; see Appendix). Studies have shown that the original 20-item scale has sound psychometric properties (e.g. Conger et al., 2000; Conger, Kanungo, Menon, & Mathur, 1997). We selected five items to cover both task- and person-focused functions (Morgeson, DeRue, & Karam, 2010).⁶ The items were referred to the

⁵ We excluded the guard and task coordinator activities. Guard activities limit the information flow across the team boundary, resulting in less boundary spanning; therefore, they are commonly omitted when boundary spanning is examined. The task coordinator activities did not fit our context as they focus on lateral interactions with actors from other departments in a larger organisation in order to pursue shared goals with other key actors (Marrone, 2010); this is not (yet) relevant in pre-founding entrepreneurial teams. To examine the validity of our short measure, we recruited a comparable sample from the same entrepreneurial training and coaching programme ($n = 63$ individuals). In this sample, we used a 16-item measure covering both ambassador and scout activities (Ancona & Caldwell, 1992). Cronbach's α for this measure was .85. A 7-item composite measure correlated very highly and significantly with the 16-item measure, $r = .90$, $p < .001$. In other words, the shortened measure consisting of seven items still represents 81 per cent of the core variance of the construct.

⁶ To examine the validity of our short measure, we recruited a comparable sample from the same entrepreneurial training and coaching programme ($n = 20$ teams). In this sample, we used the 20-item Conger-Kanungo scale to measure charismatic team leadership. We deleted the item "... is an exceptional public speaker" because this item did not fit with our context. Individual-level Cronbach's α for the 19-item scale was .96. Our 5-item aggregate measure correlated very highly and significantly with the 19-item aggregate measure, $r = .97$, $p < .001$. In other words, the shortened measure consisting of five items still represents 94 per cent of the core variance of the construct.

team leader instead of a supervisor and provided in the past tense. Followers rated the items on a Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Individual-level reliability was Cronbach's $\alpha = .70$. To legitimise data aggregation, we calculated r_{wg} , $ICC(1)$, and $ICC(2)$ as indicators for within-team agreement and between-team variance and tested for significance of the $ICC(1)$ values. The mean r_{wg} was .79 (*Median* = .87), which represents strong agreement (LeBreton & Senter, 2008). The $ICC(1)$ was .25, $F(1, 57) = 1.81$, $p < .01$, and the $ICC(2)$ was .45. $ICC(1)$ values are typically interpreted as a measure of effect size (small effect $< .10$, medium effect $< .25$, large effect $> .25$; Bliese, Halverson, & Schriesheim, 2002). Scholars have suggested that despite low ICC values, high within-team agreement ($r_{wg} > .70$) and between-team variability (significant F statistics) justify data aggregation (Bliese 2000; Klein & Kozlowski, 2000). We conclude that aggregation of individual ratings was legitimate.⁶ Charismatic team leadership was calculated using the average of follower ratings. Team-level reliability for charismatic team leadership was Cronbach's $\alpha = .73$.

Performance Measures. As teams in the pre-founding phase are required to learn how to function as a team and also to develop each individual team member's entrepreneurial capabilities, we included a team-level performance measure as well as an individual-level performance measure. Previous research shows a positive relationship between the quality of a business plan and new venture success at later phases (Foo, Wong, & Ong, 2005; Frese, Krauss, Keith, Escher, Grabarkiewicz, Luneng, Heers, Unger, & Friedrich, 2007; Shepherd, Zacharakis, & Baron, 2003), and teams are required to work closely together to develop their business model. Thus, we used the *quality of business planning* as a performance measure on the team-level. At the same time, investors base their evaluation of a new venture on an overall picture that usually includes oral presentations (see Baron & Markman, 2003; Baum, Locke, & Smith, 2001). Oral presentations transport intangible factors such as passion and the credibility of each individual team member, which investors are trying to assess before an investment decision is made (Grégoire, deKoning, & Oviatt, 2008). We therefore included *individual performance in a business pitch* as an individual-level performance measure.

- *Team-level performance measure: quality of business planning.* The written business plans were rated by an entrepreneurship expert using a Likert scale ranging from 1.0 (*very poor*) to 10.0 (*outstanding*). This assessment was based on predefined evaluation criteria. A second expert evaluated 30 per cent of the business plans. The experts' judgments were significantly correlated, $r = .79$, $p < .001$. Interrater

reliability computed as intraclass correlation was .77, $p < .001$, indicating strong agreement (LeBreton & Senter, 2008).⁷

- *Individual-level performance measure: individual performance in a business pitch.* After submission of the business plan, the team members pitched the business idea before two entrepreneurship experts. Each team member presented one part of the business model followed by a question and answer session. The expert panel evaluated individual performance during the business pitch using a 10-point Likert scale ranging from 1.0 (*very poor*) to 10.0 (*outstanding*). This assessment was based on structure and argumentation of the presentation, focusing on the question of how confidently and convincingly each team member presented the business idea. The panel was instructed to account for even small differences between the team members.

Control Variables. We carefully considered variables that may confound the proposed relationships (Bernerth & Aguinis, 2016; Spector & Brannick, 2011). We considered team size because it was shown to influence performance (Jin et al., 2017). Team size is also likely to influence team reflexivity (Weiss & Hoegl, 2015) and team members' motivation to engage in boundary spanning behaviour (Marrone et al., 2007). The correlations of team size with the study variables were not significant (see Table 1). We ascertained age and gender of the lead entrepreneur because both can leverage the effects of leadership (Kearney, 2008) by serving as "diffuse status cues" in the legitimisation of influence attempts (Jung, Vissa, & Pich, 2015). Although existing theory and previous empirical evidence support inclusion of these variables, differences were limited between groups, and the age and gender of the lead entrepreneur did not correlate with other variables (see Table 1). Therefore, we did not include the controls in hypothesis testing but present additional models with control variables to increase comparability with previous research (Spector & Brannick, 2011; see Tables 2 and 3).

⁷ To further validate the quality of the business plan as an outcome measure, we hand-collected information about the teams using archival data from the entrepreneurship and innovation centre as well as from the Internet. Of 23 teams, whose business plans were evaluated very positively (9.0 or 10.0), seven were granted access to advanced training and coaching sessions, nine won awards in business competitions and/or received funding, and three pursued activities to start a business. In contrast, of the 35 teams whose business plans were evaluated as 8.0 or worse, only one was granted access to advanced training and coaching sessions, only one was awarded/received funding, and two pursued activities to start a business.

TABLE 1
Descriptives and Intercorrelations of Study Variables on the Team Level

Variable	M	SD	Range	1	2	3	4	5	6	7	8
<i>Controls</i>											
1. Team size	3.64	00.58	3–5								
2. Age of the team leader	24.25	3.07	20–32	.10							
3. Gender of the team leader	0.24	0.43	0; 1								
<i>Team-level variables</i>											
4. Charismatic team leadership	3.55	0.57	1–5								
5. Team reflexivity	3.59	0.46	1–5								
6. Team performance	8.19	1.19	1–10								
<i>Individual-level variables</i>											
7. Boundary spanning behaviour	2.85	0.58	1–5								
8. Individual performance	8.12	1.20	1–10								

Note: $N = 58$ teams. Age of the team leader was available for $N = 55$ teams only. Gender: 0 = male, 1 = female. Correlations are given using pairwise exclusion. Individual-level variables were aggregated to the team level.

$^{\circ} p < .10$, $^* p < .05$, $^{**} p < .01$, $^{***} p < .001$ (two-tailed test).

TABLE 2
Results of OLS Regression Analysis to Test Team-Level Relationships

	Regression coefficient			Standardised regression coefficient	
	B	SE	p	β	p
Charismatic team leadership > Team performance (c')	-0.01	0.30	.969	-.01	.969
Team reflexivity > Team performance (b)	1.19	0.38	.002	.46	.002
Charismatic team leadership > Team reflexivity (a)	0.46	0.09	<.001	.57	<.001
	B	LL95CI ^a	UL95CI ^a	β	p
Total effect (c)	0.54	0.30	1.07	.26	.049
Direct effect (c')	-0.01	-0.53	0.55	-.01	.969
Indirect effect (ab)	0.55	0.12	1.06	.27	.007

Note: Level-2 $N = 58$. ^a Bias-corrected and accelerated 95% confidence intervals obtained from bootstrapping (5,000 resamples).

RESULTS

Table 1 shows the means, standard deviations, and zero-order correlations of the study variables. Our research model considers team reflexivity as a team-level process and boundary spanning behaviour as an individual-level process shaping entrepreneurial performance. Furthermore, it investigates charismatic team leadership as an antecedent of both processes and proposes indirect effects of charismatic team leadership on entrepreneurial performance at multiple levels (see Figure 1). As team-level effects were shown to be larger than individual-level effects (DeGroot et al., 2000), we first examined the hypothesised team-level relationships before testing the multilevel relationships.

Test of Team-Level Relationships

We hypothesised that team reflexivity is positively related to team performance (H1) and that charismatic team leadership predicts team reflexivity (H4), thereby influencing team performance through an indirect effect (H5). To test the proposed relationships, we used the SPSS Macro for Simple Mediation provided by Preacher and Hayes (2004). Table 2 shows the unstandardised and standardised regression coefficients for the proposed relationships.

We found a significant effect of team reflexivity on team performance when controlling for charismatic team leadership ($\beta = .46, p = .002$). Charismatic team leadership and team reflexivity were positively related ($\beta = .57, p < .001$).

TABLE 3
Results of Multilevel Linear Modelling for Individual Performance

<i>Dependent variable</i>	<i>Boundary spanning behaviour Model 1</i>	<i>Model 2</i>	<i>Individual performance Model 3a</i>	<i>Model 3b</i>
Fixed effects	1.43 (0.40)*	6.28 (0.77)*	3.80 (1.25)**	3.91 (1.98)*
Constant				
Controls				
Team size				0.36 (0.26)
Age of the team leader				-0.04 (0.06)
Gender of the team leader				-0.25 (0.44)
Level 1				
Charismatic team leadership	0.57 (0.12)*	-0.07 (0.08)	-0.15 (0.10)	-0.14 (0.11)
Boundary spanning behaviour			0.06 (0.07)	0.05 (0.06)
Team reflexivity			0.10 (0.11)	0.12 (0.11)
Level 2				
Charismatic team leadership	0.40 (0.11)*	0.52 (0.21)*	-0.03 (0.22)	-0.11 (0.25)
Boundary spanning behaviour			0.28 (0.25)	0.30 (0.27)
Team reflexivity			1.01 (0.37)**	0.99 (0.44)**
Random effects				
Level 1				
Intercept	0.33 (0.09)	0.93 (1.29)	0.93 (1.15)	0.93 (1.22)

Note: Minimum number of Level-1 $N = 124$ (follower only); Minimum number of Level-2 $N = 53$; Gender: 0 = male, 1 = female; Unstandardised estimates are reported with robust standard errors in parentheses. Model 1: Boundary spanning behaviour predicted by charismatic team leadership. Model 2: Individual performance predicted by charismatic team leadership. Model 3: Individual performance predicted by charismatic team leadership, team reflexivity, and boundary spanning behaviour (3a without/3b with control variables).

* $p < .05$, ** $p < .01$, * $p < .001$ (two tailed test).

The total effect of charismatic team leadership on team performance was significant ($\beta = .26$, $p = .049$), whereas the direct effect was minimised to a null-correlation when introducing the mediator team reflexivity ($\beta = -.01$, $p = .969$). The indirect effect was judged to be significant using the Sobel test (1982), $\beta = .27$, $z = 2.677$, $p = .007$. For the indirect effect, the bias-corrected and accelerated confidence interval obtained from bootstrapping (5,000 resamples) revealed a lower limit of 0.12 and an upper limit of 1.06, not including zero (Shrout & Bolger, 2002). In short, our data provided support for the assumption that team reflexivity is related to team performance (H1) and that charismatic team leadership predicts team reflexivity (H4), ultimately affecting team performance through a team-level mediation (H5).

Test of Multilevel Relationships

We hypothesised that boundary spanning behaviour is related to individual performance (H2) and that team reflexivity exerts cross-level effects on individual performance (H3). Based on the finding that charismatic team leadership facilitated team reflexivity, we tested an indirect effect of charismatic team leadership on individual performance through team reflexivity (2–2–1 mediation; H6). We further suggested that charismatic team leadership is related to boundary spanning behaviour (H7) and that it exerts indirect effects on individual performance through boundary spanning behaviour (2–1–1 mediation; H8). Multilevel modelling was necessary and appropriate to test the proposed relationships (Raudenbush & Bryk, 2002) as individual performance varied significantly between teams, $\chi^2(55) = 867.958$, $p < .001$. We used HLM 7.0 with restricted maximum-likelihood estimation for data analysis. Following the recommendation of Mathieu and Taylor (2007), we tested within-level relationships before investigating cross-level relationships. The analyses used group-mean centred level-1 variables and uncentred level-2 variables (Zhang, Zyphur, & Preacher, 2009). To be able to estimate the distinct effects of our study variables as they are conceptualised at their respective levels of analysis, we introduced the individual ratings of charismatic team leadership and team reflexivity as level-1 covariates and an aggregated measure of individual boundary spanning behaviour as level-2 covariate (Zhang et al., 2009). Table 3 presents the multilevel linear modelling results.

Model 1 shows that charismatic team leadership was related to individual boundary spanning behaviour, $\gamma = 0.40$, $p < .001$. The significant effect of charismatic team leadership on team reflexivity was detailed above. The results of Model 2 show that charismatic team leadership predicted individual performance, $\gamma = 0.52$, $p = 0.016$. In order to establish the proposed multilevel mediation, we predicted individual performance from charismatic team leadership while introducing the proposed level-2 mediator, team reflexivity, and the proposed level-1 mediator, boundary spanning behaviour. Following Zhang et al.'s (2009) approach to differentiate mediation effects on different levels, we introduced the level-2 predictors as level-1 covariate and vice versa (see Model 3). The results of Model 3 revealed that individual boundary spanning behaviour was not related to individual performance, $\gamma = 0.06$, $p = .381$, whereas team reflexivity was, $\gamma = 1.01$, $p = .009$. In this model, there was no direct effect of charismatic team leadership on individual performance, $\gamma = 0.03$, $p = .907$, thus supporting the cross-level mediation through team reflexivity. The indirect effect of charismatic team leadership on individual performance through team reflexivity was confirmed by the Sobel (1982) test, $z = 2.403$, $p = .016$. We conclude that team reflexivity (H3)—but not boundary spanning behaviour (H2)—was related to individual performance. Although both were predicted by charismatic team leadership as hypothesised (H4, H7), the effect of

TABLE 4
Summary of Empirical Results

<i>Hypothesis</i>	<i>Result</i>
<i>Effects of team reflexivity and boundary spanning behaviour on performance</i>	
H1: Team reflexivity is positively related to entrepreneurial team performance.	Supported
H2: Boundary spanning behaviour is positively related to the performance of individual members of the entrepreneurial team.	Not supported
H3: Team reflexivity is positively related to the performance of individual members of the entrepreneurial team.	Supported
<i>Indirect effects of charismatic leadership via team reflexivity</i>	
H4: Charismatic team leadership is positively related to team reflexivity.	Supported
H5: Charismatic team leadership is positively related to entrepreneurial team performance and this relationship is mediated by team reflexivity.	Supported
H6: Charismatic team leadership is positively related to the performance of individual members of the entrepreneurial team and this relationship is mediated by team reflexivity.	Supported
<i>Indirect effects of charismatic leadership via boundary spanning behaviour</i>	
H7: Charismatic team leadership is positively related to boundary spanning behaviour of members of the team.	Supported
H8: Charismatic team leadership is positively related to the performance of individual members of the entrepreneurial team and this relationship is mediated by boundary spanning behaviour.	Not supported

charismatic team leadership on individual performance was mediated by team reflexivity (H6), but not by boundary spanning behaviour (H8). Table 4 provides a summary of the hypotheses testing.

DISCUSSION

The pre-founding phase is highly demanding for entrepreneurial teams and for each member of the team, and a team's performance at this stage will have a long-term impact on new venture success (Fern et al., 2012). Pre-founding teams often struggle with multiple simultaneous challenges concerning both the task (i.e. recognising business opportunities, selecting the most promising idea, and constantly refining the business model while changes in all facets of the nascent venture occur) and the team (i.e. establishing team functioning and expanding team members' entrepreneurial capabilities while the team is still forming). The way in which teams shape their teamwork is key in leveraging performance at this stage (Bird et al., 2012; Cohen & Bailey, 1997). In our research, we explored the contribution of two factors in facilitating performance in the light of the challenges associated with the pre-founding phase:

learning and leadership. First, team learning is key in refining or pivoting business ideas, in learning from experience to optimise teamwork, and in evaluating progress towards the team's vision (Keith et al., 2016). At the same time, individual learning behaviours seem crucial because each member of the team needs to develop his or her entrepreneurial capabilities (Gielnik et al., 2014). Second, team leadership is needed that facilitates the development of the team's business from a vague idea into a shared vision, encourages out-of-the-box-thinking of the whole team and its members, and enables everybody to excel (Hmielski & Ensley, 2007).

The research reported in this paper provided evidence for the viability of a multilevel model of learning and leadership in predicting entrepreneurial performance in pre-founding teams. Specifically, our study is the first to investigate team reflexivity and individual boundary spanning behaviour as drivers of entrepreneurial performance in pre-founding teams. We hypothesised and found that team reflexivity enables teams to adapt their teamwork to the rapidly changing and highly volatile pre-founding phase, thereby shaping the performance of the whole team and of each individual team member. Contrary to our hypothesis, boundary spanning behaviour was not related to each team member's individual entrepreneurial performance. Charismatic team leadership was found to facilitate team reflexivity and individual boundary spanning behaviour, providing new insights into how leadership shapes processes and outcomes of entrepreneurial teams.

Theoretical Implications

Our findings make important contributions in advancing our understanding of the facilitators that leverage performance at a very early stage of the entrepreneurial endeavour—the pre-founding phase. They add to the literature on entrepreneurial teams in two ways. First, we expand the dominant models of entrepreneurial success that focused on input factors, largely neglecting process factors (for a review see Jin et al., 2017). Although we acknowledge the significance of input factors such as team composition characteristics, our research points to the importance of *team processes* in shaping entrepreneurial team performance (see also Kamm, Shuman, Seeger, & Nurick, 1990). Specifically, we argued that learning processes seem to be of particular relevance in the pre-founding phase (Ravasi & Turati, 2005; Wang & Chugh, 2014). We contribute to the advancement of research on entrepreneurial learning by highlighting the role of team reflexivity, that is, the continuous evaluation and discussion of team goals and means to achieve these goals. In doing so, we respond to earlier calls to examine *the team as the agent of learning* (Wang & Chugh, 2014). Our findings suggest that simply putting a team together does not imply that this team is able to act effectively in the highly volatile and unpredictable environment of the pre-founding phase; instead, only reflexive teams seem to be

readily prepared to face the task and team challenges associated with the early stages of the entrepreneurial endeavour.

As a previously neglected individual learning activity, we analysed each individual team member's boundary spanning behaviour. Our findings imply that charismatic leadership encourages boundary spanning behaviour and augment previous theorising on the relevance of leadership for crossing team boundaries (DeChurch & Marks, 2006; Joshi, Pandey, & Han, 2009). However, boundary spanning behaviour was not related to individual performance in our study. As Marrone et al. (2007) found that team members may suffer from role overload when challenged to balance internal and external activities, a plausible reason for this finding relates to the small team size in our research; acting in a small team has likely put heightened demands on each member of the team, resulting in detrimental effects on individual performance in some cases. This finding challenges the positivist view on boundary spanning behaviour, pointing to the significance of moderators (Faraj & Yan, 2009; Marrone, 2010).

Second, we contribute to the literature by investigating how leadership impacts learning processes in the pre-founding phase, responding to calls to explore the role of the team leader in entrepreneurial teams (Antonakis & Auto, 2007). This perspective is needed as "all entrepreneurs are by definition leaders" (Ensley et al., 2006, p. 246). Leadership is one of the most critical factors in the success of any entrepreneurial team in facilitating and sustaining collective and individual performance in entrepreneurship (Hmielski & Ensley, 2007), "making new ventures a particularly important context in which to examine the effects of leadership behavior" (Ensley et al., 2006, p. 219). In identifying opportunities in the environment, communicating a clear vision, and empowering all members of the team to achieve this vision, charismatic team leadership of the lead entrepreneur was expected to foster both team performance and individual performance. Our findings imply that charismatic team leadership is effective in shaping both team reflexivity and boundary spanning behaviour, addressing calls to investigate the role of leaders in influencing entrepreneurial teams to perform better (Gorgievski & Stephan, 2016). Moreover, to our knowledge, this study is the first to examine team reflexivity and boundary spanning behaviour as multilevel pathways of the effect of charismatic leadership. Although warranting replication, our pattern of results highlights cross-level effects of leadership flowing from the team to the individual level, which have largely been ignored in favour of within-level effects (DeChurch, Hiller, Murase, Doty, & Salas, 2010; Lord & Dinh, 2012).

Beyond work on entrepreneurial teams, this study also extends research on the significance of leadership in initiating and facilitating team reflexivity, which was called for by Schippers et al. (2008). This perspective is particularly relevant as teams in general are not very reflexive (West, 2000). We theorise

and find that charismatic team leadership fosters team reflexivity, enabling teams and individual members of the team to perform better. This finding challenges the assumption that charismatic leadership increases group-think phenomena (Klein & House, 1995). In contrast, the joint vision put forward by a charismatic leader seems to function as a shared reference to critically evaluate progress towards the team goals (Schippers et al., 2008).

Limitations and Future Research

This study broadly contributes to the literature; however, it is not without limitations. First, our research model was restricted to the examination of charismatic team leadership on the team level. Individuals may vary with regard to the quality of their relationship with the team leader (Kozlowski et al., 2016). For example, leader-member-exchange (LMX) theory assumes that leaders develop individual relationships with their followers (Schriesheim, Castro, & Coglisier, 1999). Similarly, team reflexivity was examined as a collective team phenomenon in our study. Still, each team member's *individual* reflexivity on their own may well contribute to team reflexivity in terms of a bottom-up process (e.g. Gong, Kim, Lee, & Zhu, 2013). We suggest future research to include LMX and individual reflexivity as predictors of performance.

A clear strength of our study is the field setting; still, we cannot rule out reverse causality. Schippers et al. (2008) found that prior team performance affects team reflexivity. Moreover, charismatic team leadership ratings may depend on perceived team performance and causal ascription of performance to the team leader (Lord, Brown, Harvey, & Hall, 2001; Weber, Camerer, Rottenstreich, & Knez, 2001). Although the performance evaluations were only available to the participants after we assessed the variables of our research and the outcome measures were based on expert assessments, our findings warrant replication using longitudinal designs. Due to the field setting, we used short measures in order to maximise response rates. In a comparable sample, the measures used in this research correlated very highly with the original scales, which indicates that the core variance of the focal constructs under investigation was captured. Still, we encourage researchers to replicate our findings with the full scales provided by Conger et al. (2000), Schippers et al. (2007), and Ancona and Caldwell (1992).

Finally, we recruited a sample of rather homogeneous teams from a venture creation programme. This allowed us to investigate how entrepreneurial teams shape their teamwork in unique ways under very comparable conditions. We are confident that our findings—with some caution—can be generalised to other populations of teams as long as members of the team work together on a highly motivated and cooperative basis in a dynamic and complex environment characterised by the elevated importance of innovation.

REFERENCES

- Agle, B.R., Nagarajan, N.J., Sonnenfeld, J.A., & Srinivasan, D. (2006). Does CEO charisma matter? An empirical analysis of the relationships among organizational performance, environmental uncertainty, and top management team perceptions of CEO charisma. *Academy of Management Journal*, 49, 161–174.
- Ancona, D.G., & Caldwell, D.F. (1992). Bridging the boundary: External activity and performance in organizational teams. *Administrative Science Quarterly*, 37, 634–665.
- Antonakis, J., & Autio, E. (2007). Entrepreneurship and leadership. In J. R. Baum, M. Frese, & R. Baron (Eds.), *The psychology of entrepreneurship* (pp. 1–10). London: Routledge.
- Ardichvili, A., Cardozo, R., & Ray, S. (2003). A theory of entrepreneurial opportunity identification and development. *Journal of Business Venturing*, 18, 105–123.
- Bae, T.J., Qian, S., Miao, C., & Fiet, J.O. (2014). The relationship between entrepreneurship education and entrepreneurial intentions: A meta-analytic review. *Entrepreneurship Theory and Practice*, 38, 217–254.
- Baron, R.A., & Markman, G.D. (2003). Beyond social capital: The role of entrepreneurs' social competence in their financial success. *Journal of Business Venturing*, 18, 41–60.
- Baron, R.A., & Tang, J. (2009). Entrepreneurs' social skills and new venture performance: Mediating mechanisms and cultural generality. *Journal of Management*, 35, 282–306.
- Baum, J.R., Locke, E.A., & Kirkpatrick, S.A. (1998). A longitudinal study of the relation of vision and vision communication to venture growth in entrepreneurial firms. *Journal of Applied Psychology*, 83, 43–54.
- Baum, J.R., Locke, E.A., & Smith, K.G. (2001). A multi-dimensional model of venture growth. *Academy of Management Journal*, 44, 292–303.
- Bernerth, J.B., & Aguinis, H. (2016). A critical review and best-practice recommendations for control variable usage. *Personnel Psychology*, 69, 229–283.
- Bird, B., Schjoedt, L., & Baum, J.R. (2012). Editor's introduction. Entrepreneurs' behavior: Elucidation and measurement. *Entrepreneurship Theory and Practice*, 36, 889–913.
- Blackburn, R., & Kovalainen, A. (2009). Researching small firms and entrepreneurship: Past, present and future. *International Journal of Management Reviews*, 11, 127–148.
- Bliese, P.D. (2000). Within-group agreement, non-independence, and reliability: Implications for data aggregation and analysis. In K.J. Klein & S.W.J. Kozlowski (eds.), *Multilevel theory, research, and methods in organizations: Foundations, extensions, and new directions* (pp. 349–381). San Francisco, CA: Jossey-Bass.
- Bliese, P.D., Halverson, R.R., & Schriesheim, C.A. (2002). Benchmarking multilevel methods in leadership: *The articles, the model, and the data set*. *The Leadership Quarterly*, 13, 3–14.
- Braun, S.*, Peus, C.*, Weisweiler, S., & Frey, D. (2013). Transformational leadership, job satisfaction, and team performance: A multilevel mediation model of trust. *The Leadership Quarterly*, 24, 270–283. [*equal contribution]
- Cardon, M.S., Zietsma, C., Saparito, P., Matherne, B.P., & Davis, C. (2005). A tale of passion: New insights into entrepreneurship from a parenthood metaphor. *Journal of Business Venturing*, 20, 23–45.

- Chan, D. (1998). Functional relations among constructs in the same content domain at different levels of analysis: A typology of composition models. *Journal of Applied Psychology*, 83, 234–246.
- Cogliser, C.C., & Brigham, K.H. (2004). The intersection of leadership and entrepreneurship: Mutual lessons to be learned. *The Leadership Quarterly*, 15, 771–799.
- Cohen, S.G., & Bailey, D.E. (1997). What makes teams work: Group effectiveness from the shop floor to the executive suite. *Journal of Management*, 23, 239–290.
- Conger, J.A., Kanungo, R.N., & Menon, S.T. (2000). Charismatic leadership and follower effects. *Journal of Organizational Behavior*, 21, 747–767.
- Conger, J.A., Kanungo, R.N., Menon, S.T., & Mathur, P. (1997). Measuring charisma: Dimensionality and validity of the Conger–Kanungo scale of charismatic leadership. *Canadian Journal of Administrative Sciences*, 14, 290–302.
- Cope, J. (2003). Entrepreneurial learning and critical reflection: Discontinuous events as triggers for higher-level learning. *Management Learning*, 34, 429–450.
- Crossan, M.M., Lane, H.W., & White, R.E. (1999). An organizational learning framework: From intuition to institution. *Academy of Management Review*, 24, 522–537.
- Davidsson, P., & Wiklund, J. (2001). Levels of analysis in entrepreneurship research: Current research practice and suggestions for the future. *Entrepreneurship Theory & Practice*, 25(4), 81–100.
- DeChurch, L.A., & Marks, M.A. (2006). Leadership in multiteam systems. *Journal of Applied Psychology*, 91, 311–329.
- DeChurch, L.A., Hiller, N.J., Murase, T., Doty, D., & Salas, E. (2010). Leadership across levels: Levels of leaders and their levels of impact. *The Leadership Quarterly*, 21, 1069–1085.
- DeDreu, C.K. (2002). Team innovation and team effectiveness: The importance of minority dissent and reflexivity. *European Journal of Work and Organizational Psychology*, 11, 285–298.
- DeGroot, T., Kiker, D.S., & Cross, T.C. (2000). A meta-analysis to review organizational outcomes related to charismatic leadership. *Canadian Journal of Administrative Sciences*, 17, 356–372.
- DeMol, E., Khapova, S.N., & Elfring, T. (2015). Entrepreneurial team cognition: A review. *International Journal of Management Reviews*, 17, 232–255.
- Ensley, M.D., Carland, J.W., & Carland, J.C. (2000). Investigating the existence of the lead entrepreneur. *Journal of Small Business Management*, 38, 59–77.
- Ensley, M.D., Pearce, C.L., & Hmieleski, K.M. (2006). The moderating effect of environmental dynamism on the relationship between entrepreneur leadership behavior and new venture performance. *Journal of Business Venturing*, 21, 243–263.
- Faraj, S., & Yan, A. (2009). Boundary work in knowledge teams. *Journal of Applied Psychology*, 94, 604–617.
- Fern, M.J., Cardinal, L.B., & O'Neill, H.M. (2012). The genesis of strategy in new ventures: Escaping the constraints of founder and team knowledge. *Strategic Management Journal*, 33, 427–447.
- Foo, M.D., Sin, H.P., & Yiong, L.P. (2006). Effects of team inputs and intrateam processes on perceptions of team viability and member satisfaction in nascent ventures. *Strategic Management Journal*, 27(4), 389–399.

- Foo, M.D., Wong, P.K., & Ong, A. (2005). Do others think you have a viable business idea? Team diversity and judges' evaluation of ideas in a business plan competition. *Journal of Business Venturing*, 20, 385–402.
- Frese, M., Chell, E., & Klandt, H. (2000). Introduction. *European Journal of Work and Organizational Psychology*, 9, 3–6.
- Frese, M., Krauss, S.I., Keith, N., Escher, S., Grabarkiewicz, R., Luneng, S.T., Heers, C., Unger, J., & Friedrich, C. (2007). Business owners' action planning and its relationship to business success in three African countries. *Journal of Applied Psychology*, 92, 1481–1498.
- Fulmer, C.A., & Ostroff, C. (2016). Convergence and emergence in organizations: An integrative framework and review. *Journal of Organizational Behavior*, 37, 122–145.
- Gielnik, M.M., Krämer, A.C., Kappel, B., & Frese, M. (2014). Antecedents of business opportunity identification and innovation: Investigating the interplay of information processing and information acquisition. *Applied Psychology*, 63, 344–381.
- Gong, Y., Kim, T.Y., Lee, D.R., & Zhu, J. (2013). A multilevel model of team goal orientation, information exchange, and creativity. *Academy of Management Journal*, 56, 827–851.
- Gorgievski, M.J., & Stephan, U. (2016). Advancing the psychology of entrepreneurship: A Review of the psychological literature and an introduction. *Applied Psychology*, 65, 437–468.
- Grégoire, D.A., DeKoning, A.J., & Oviatt, B.M. (2008). Do VCs evaluate “live” presentations like they evaluate business plans? *Frontiers of Entrepreneurship Research*, 28(3), 1–13.
- Hirst, G., & Mann, L. (2004). A model of R&D leadership and team communication: the relationship with project performance. *R&D Management*, 34(2), 147–160.
- Hmieleski, K.M., & Ensley, M.D. (2007). A contextual examination of new venture performance: Entrepreneur leadership behavior, top management team heterogeneity, and environmental dynamism. *Journal of Organizational Behavior*, 28, 865–889.
- Hoegl, M., & Parboteeah, K.P. (2006). Team reflexivity in innovative projects. *R&D Management*, 36, 113–125.
- Hollingshead, A., & Poole, M.S. (Eds.). (2012). *Research methods for studying groups and teams: A guide to approaches, tools, and technologies*. New York: Routledge.
- Ireland, R.D., & Webb, J.W. (2007). A cross-disciplinary exploration of entrepreneurship research. *Journal of Management*, 33, 891–927.
- Jacobsen, C., & House, R.J. (2001). Dynamics of charismatic leadership: A process theory, simulation model, and tests. *The Leadership Quarterly*, 12, 75–112.
- Jin, L., Madison, K., Kraiczy, N.D., Kellermanns, F.W., Crook, T.R., & Xi, J. (2017). Entrepreneurial team composition characteristics and new venture performance: A meta-analysis. *Entrepreneurship Theory and Practice*, 41, 743–771.
- Johnson, P.S., Parker, S.C., & Wijbenga, F. (2006). Nascent entrepreneurship research: Achievements and opportunities. *Small Business Economics*, 27, 1–4.
- Joshi, A., Pandey, N., & Han, G. (2009). Bracketing team boundary spanning: An examination of task-based, team-level, and contextual antecedents. *Journal of Organizational Behavior*, 30, 731–759.

- Jung, H., Vissa, B., & Pich, M. (2015). How do entrepreneurial founding teams allocate task positions? *Academy of Management Journal*, 60(1), 264–294.
- Kamm, J.B., Schuman, J.C., Seeger, J.A., & Nurick, A.J. (1990). Entrepreneurial teams in new venture creation: A research agenda. *Entrepreneurship Theory and Practice*, 14(4), 7–17.
- Kearney, E. (2008). Age differences between leader and followers as a moderator of the relationship between transformational leadership and team performance. *Journal of Occupational and Organizational Psychology*, 81, 803–811.
- Keith, N., Unger, J.M., Rauch, A., & Frese, M. (2016). Informal learning and entrepreneurial success: A longitudinal study of deliberate practice among small business owners. *Applied Psychology*, 65, 515–540.
- Klein, K.J., & House, R.J. (1995). On fire: Charismatic leadership and levels of analysis. *The Leadership Quarterly*, 6, 183–198.
- Klein, K.J., & Kozlowski, S.W. (2000). From micro to meso: Critical steps in conceptualizing and conducting multilevel research. *Organizational Research Methods*, 3, 211–236.
- Klotz, A.C., Hmieleski, K.M., Bradley, B.H., & Busenitz, L.W. (2014). New venture teams: A review of literature and roadmap for future research. *Journal of Management*, 40, 226–255.
- Kozlowski, S.W.J., & Klein, K.J. (2000). A multi-level approach to theory and research in organizations: Contextual, temporal, and emergent processes. In K.J. Klein & S.W.J. Kozlowski (Eds.), *Multilevel theory, research, and methods in organizations: Foundations, extensions, and new directions* (pp. 3–90). San Francisco: Jossey-Bass.
- Kozlowski, S.W., Mak, S., & Chao, G.T. (2016). Team-centric leadership: An integrative review. *Annual Review of Organizational Psychology and Organizational Behavior*, 3, 21–54.
- Lackéus, M., & Williams Middleton, K. (2015). Venture creation programs: Bridging entrepreneurship education and technology transfer. *Education + Training*, 57(1), 48–73.
- LeBreton, J.M., & Senter, J.L. (2008). Answers to 20 questions about interrater reliability and interrater agreement. *Organizational Research Methods*, 11, 815–852.
- Leung, A., Zhang, J., Wong, P.K., & Foo, M.D. (2006). The use of networks in human resource acquisition for entrepreneurial firms: Multiple “fit” considerations. *Journal of Business Venturing*, 21, 664–686.
- Lord, R.G., Brown, D.J., Harvey, J.L., & Hall, R.J. (2001). Contextual constraints on prototype generation and their multilevel consequences for leadership perceptions. *The Leadership Quarterly*, 12, 311–338.
- Lord, R.G., & Dinh, J.E. (2012). Aggregation processes and levels of analysis as organizing structures for leadership theory. In D.V. Day & J. Antonakis (Eds.), *The nature of leadership* (2nd edn, pp. 29–65). Thousand Oaks, CA: Sage Publications.
- Marrone, J.A. (2010). Team boundary spanning: A multilevel review of past research and proposals for the future. *Journal of Management*, 36, 911–940.
- Marrone, J.A., Tesluk, P.E., & Carson, J.B. (2007). A multilevel investigation of antecedents and consequences of team member boundary-spanning behavior. *Academy of Management Journal*, 50, 1423–1439.

- Martin, B.C., McNally, J.J., & Kay, M.J. (2013). Examining the formation of human capital in entrepreneurship: A meta-analysis of entrepreneurship education outcomes. *Journal of Business Venturing*, 28, 211–224.
- Mathieu, J.E., & Taylor, S.R. (2007). A framework for testing meso-mediational relationships in organizational behavior. *Journal of Organizational Behavior*, 28, 141–172.
- McMullen, J.S., & Shepherd, D.A. (2006). Entrepreneurial action and the role of uncertainty in the theory of the entrepreneur. *Academy of Management Review*, 31, 132–152.
- Morgeson, F.P., DeRue, D.S., & Karam, E.P. (2010). Leadership in teams: A functional approach to understanding leadership structures and processes. *Journal of Management*, 36, 5–39.
- Nohe, C., Michaelis, B., Menges, J.I., Zhang, Z., & Sonntag, K. (2013). Charisma and organizational change: A multilevel study of perceived charisma, commitment to change, and team performance. *The Leadership Quarterly*, 24, 378–389.
- Otte, K.P., Konradt, U., Garbers, Y., & Schippers, M.C. (2017). Development and validation of the REMINT: A reflection measure for individuals and teams. *European Journal of Work and Organizational Psychology*, 26, 299–313.
- Paulsen, N., Maldonado, D., Callan, V.J., & Ayoko, O. (2009). Charismatic leadership, change and innovation in an R&D organization. *Journal of Organizational Change Management*, 22, 511–523.
- Preacher, K.J., & Hayes, A.F. (2004). SPSS and SAS procedures for estimating indirect effects in simple mediation models. *Behavior Research Methods, Instruments, and Computers*, 36, 717–731.
- Raudenbusch, S.W., & Bryk, A.S. (2002). *Hierarchical linear models: Applications and data analysis methods*. Thousand Oaks, CA: Sage.
- Ravasi, D., & Turati, C. (2005). Exploring entrepreneurial learning: A comparative study of technology development projects. *Journal of Business Venturing*, 20, 137–164.
- Schippers, M.C., Den Hartog, D.N., & Koopman, P.L. (2007). Reflexivity in teams: A measure and correlates. *Applied Psychology*, 56, 189–211.
- Schippers, M.C., Den Hartog, D.N., Koopman, P.L., & van Knippenberg, D. (2008). The role of transformational leadership in enhancing team reflexivity. *Human Relations*, 61, 1593–1616.
- Schippers, M.C., West, M.A., & Dawson, J.F. (2015). Team reflexivity and innovation: The moderating role of team context. *Journal of Management*, 41, 769–788.
- Schjoedt, L., & Kraus, S. (2009a). The heart of a new venture: The entrepreneurial team. *Management Research News*, 32, 509–512.
- Schjoedt, L., & Kraus, S. (2009b). Entrepreneurial teams: Definition and performance factors. *Management Research News*, 32, 513–524.
- Schriesheim, C.A., Castro, S.L., & Cogliser, C.C. (1999). Leader-member exchange (LMX) research: A comprehensive review of theory, measurement, and data-analytic practices. *The Leadership Quarterly*, 10, 63–113.
- Shamir, B., & Howell, J.M. (1999). Organizational and contextual influences on the emergence and effectiveness of charismatic leadership. *The Leadership Quarterly*, 10, 257–283.
- Shamir, B., House R.J., & Arthur, M.B. (1993). The motivational effects of charismatic leadership: A self-concept based theory. *Organization Science*, 4, 577–594.

- Shea, C., & Howell, J. (1999). Charismatic leadership and task feedback: A laboratory study of their effects on self-efficacy and task performance. *The Leadership Quarterly*, 10, 375–396.
- Shepherd, D.A., Zacharakis, A., & Baron, R.A. (2003). VCs' decision processes: Evidence suggesting more experience may not always be better. *Journal of Business Venturing*, 18, 381–401.
- Shin, Y., Kim, M., & Lee, S.H. (2016). Reflection toward creativity: Team reflexivity as a linking mechanism between team goal orientation and team creative performance. *Journal of Business and Psychology*. Online First.
- Shrout, P.E., & Bolger, N. (2002). Mediation in experimental and nonexperimental studies: New procedures and recommendations. *Psychological Methods*, 7, 422–445.
- Sobel, M.E. (1982). Asymptotic confidence intervals for indirect effects in structural equation models. In S. Leinhardt (Ed.), *Sociological methodology* (pp. 290–312). Washington, DC: American Sociological Association.
- Somech, A. (2006). The effects of leadership style and team process on performance and innovation in functionally heterogeneous teams. *Journal of Management*, 32, 132–57.
- Spector, P.E., & Brannick, M.T. (2011). Methodological urban legends: The misuse of statistical control variables. *Organizational Research Methods*, 14, 287–305.
- Srivastava, A., Bartol, K.M., & Locke, E.A. (2006). Empowering leadership in management teams: Effects on knowledge sharing, efficacy, and performance. *Academy of Management Journal*, 49, 1239–1251.
- Tesluk, P.E., & Mathieu, J.E. (1999). Overcoming roadblocks to effectiveness: Incorporating management of performance barriers into models of work group effectiveness. *Journal of Applied Psychology*, 84, 200–217.
- Van Mierlo, H., Vermunt, J.K., & Rutte, C.G. (2009). Composing group-level constructs from individual-level survey data. *Organizational Research Methods*, 12, 368–392.
- Wang, C.L., & Chugh, H. (2014). Entrepreneurial learning: Past research and future challenges. *International Journal of Management Reviews*, 16, 24–61.
- Wang, X.-H., & Howell, J.M. (2010). Exploring the dual-level effects of transformational leadership on followers. *Journal of Applied Psychology*, 95, 1134–1144.
- Weber, R., Camerer, C., Rottenstreich, Y., & Knez, M. (2001). The illusion of leadership: Misattribution of cause in coordination games. *Organization Science*, 12, 582–598.
- Weiss, M., & Hoegl, M. (2016). Effects of relative team size on teams with innovative tasks: An understaffing theory perspective. *Organizational Psychology Review*, 6, 324–351.
- West, M.A. (1996). Reflexivity and work group effectiveness: A conceptual integration. In M.A. West (ed.), *Handbook of work group psychology* (pp. 555–579). Chichester: John Wiley & Sons.
- West, M.A. (2000). Reflexivity, revolution and innovation in work teams. In M.M. Beyerlein, D.A. Johnson, & S.T. Beyerlein (eds.), *Product development teams* (Vol. 5, pp. 1–29). Stanford CT: JAI Press.
- Woehr, D.J., Loignon, A.C., Schmidt, P.B., Loughry, M.L., & Ohland, M.W. (2015). Justifying aggregation with consensus-based constructs: A review and

- examination of cutoff values for common aggregation indices. *Organizational Research Methods*, 18, 704–737.
- Yukl, G.A. (2002). *Leadership in organizations* (5th edn). Upper Saddle River, NJ: Prentice Hall.
- Zhang, Z., Zyphur, M.J., & Preacher, K.J. (2009). Testing multilevel mediation using hierarchical linear models: Problems and solutions. *Organizational Research Methods*, 12, 695–719.

APPENDIX: Survey measures used in this research

Charismatic team leadership (based on Conger et al., 2000)

- The team leader engaged in unconventional behaviour in order to achieve the goals of the team.
- The team leader influenced others by developing mutual liking and respect.
- The team leader often incurred high personal cost for the good of the team.
- The team leader showed sensitivity for the needs and feelings of the other members of the team.
- The team leader often exhibited very unique behaviour that surprised other members of the team.

Team reflexivity (based on Schippers et al., 2007)⁸

- We worked out what we can learn from past experience.
- During the task execution, we stopped to assess whether the team was on the right track.
- We regularly checked whether our activities had led to the expected results.
- If things didn't work out as planned, we discussed the possible causes for that.
- We regularly discussed our methods for getting the job done.
- We often discussed what we could learn from our failures and mistakes.
- We regularly discussed whether the team was working effectively.
- We evaluated our project progress on a regular basis.

Boundary spanning behaviour (based on Ancona & Caldwell, 1992)

- Talk up the team to outsiders.
 - Report the progress of the team to others.
 - Keep others informed of your team's activities.
 - Find out what competing groups are doing on similar projects.
 - Scan the environment for marketing ideas/expertise.
 - Scan the environment for technical ideas/expertise.
 - Protect the team from outside interference
-

⁸ Items 6 and 8 were slightly adapted with regard to wording. These adaptations were based on the theoretical considerations put forward by Schippers et al. (2007) and to make more explicit the aspects of learning from mistakes and monitoring of the progress towards team goals. Original wording was: "If things did not work out as planned, we considered what to do about it" and "The team often reviewed whether it was getting the job done". Team reflexivity measured only by the six items that were not adapted from Schippers et al. (2007), that is items 1 to 5 and 7, correlated highly significantly with the 8-item measure, $r = .97$, $p < .001$. We conclude that the adaptation of the items 6 and 8 did not impact validity of the team reflexivity measure.